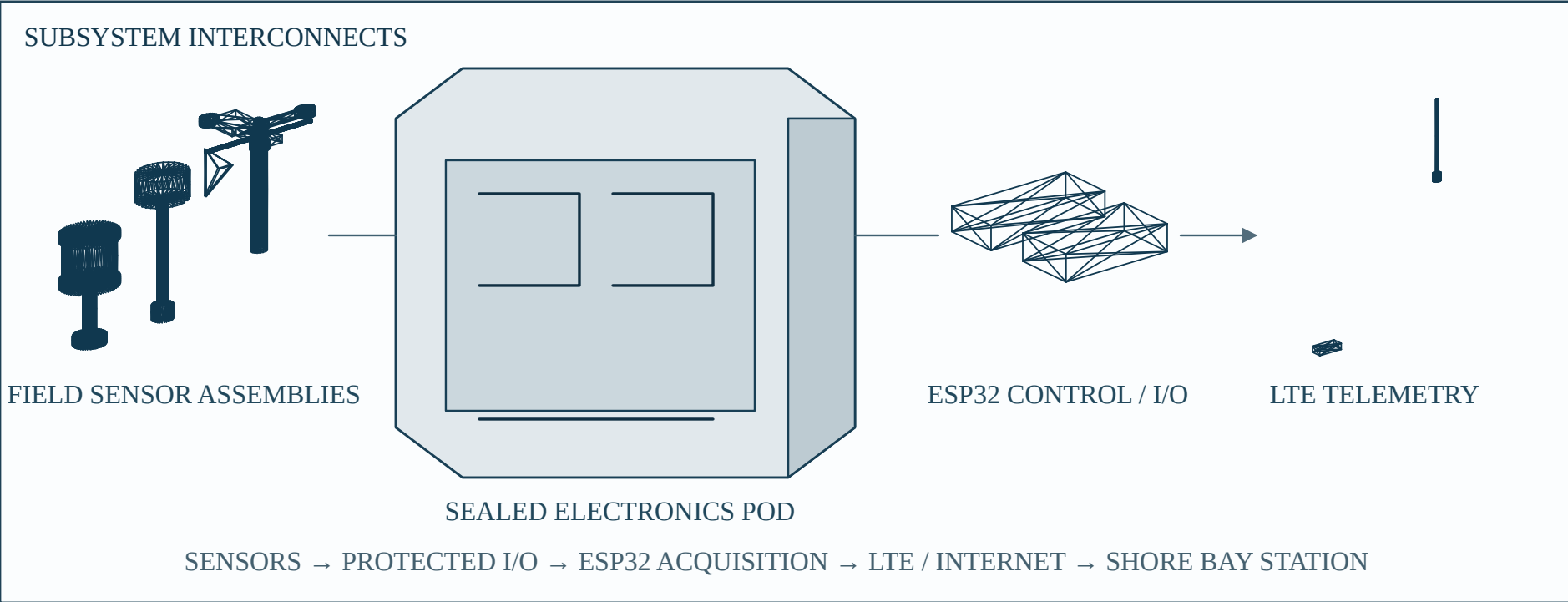
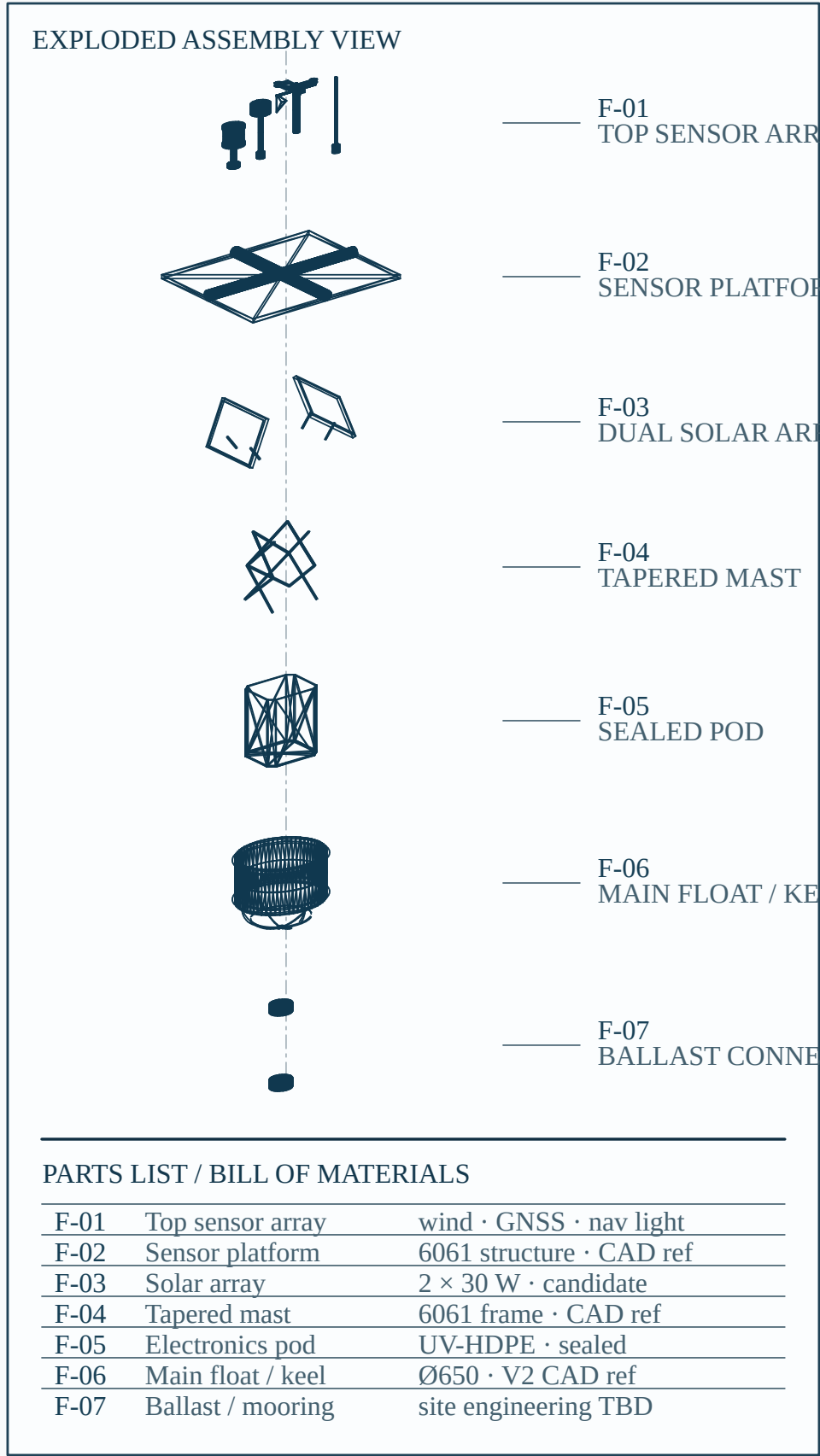
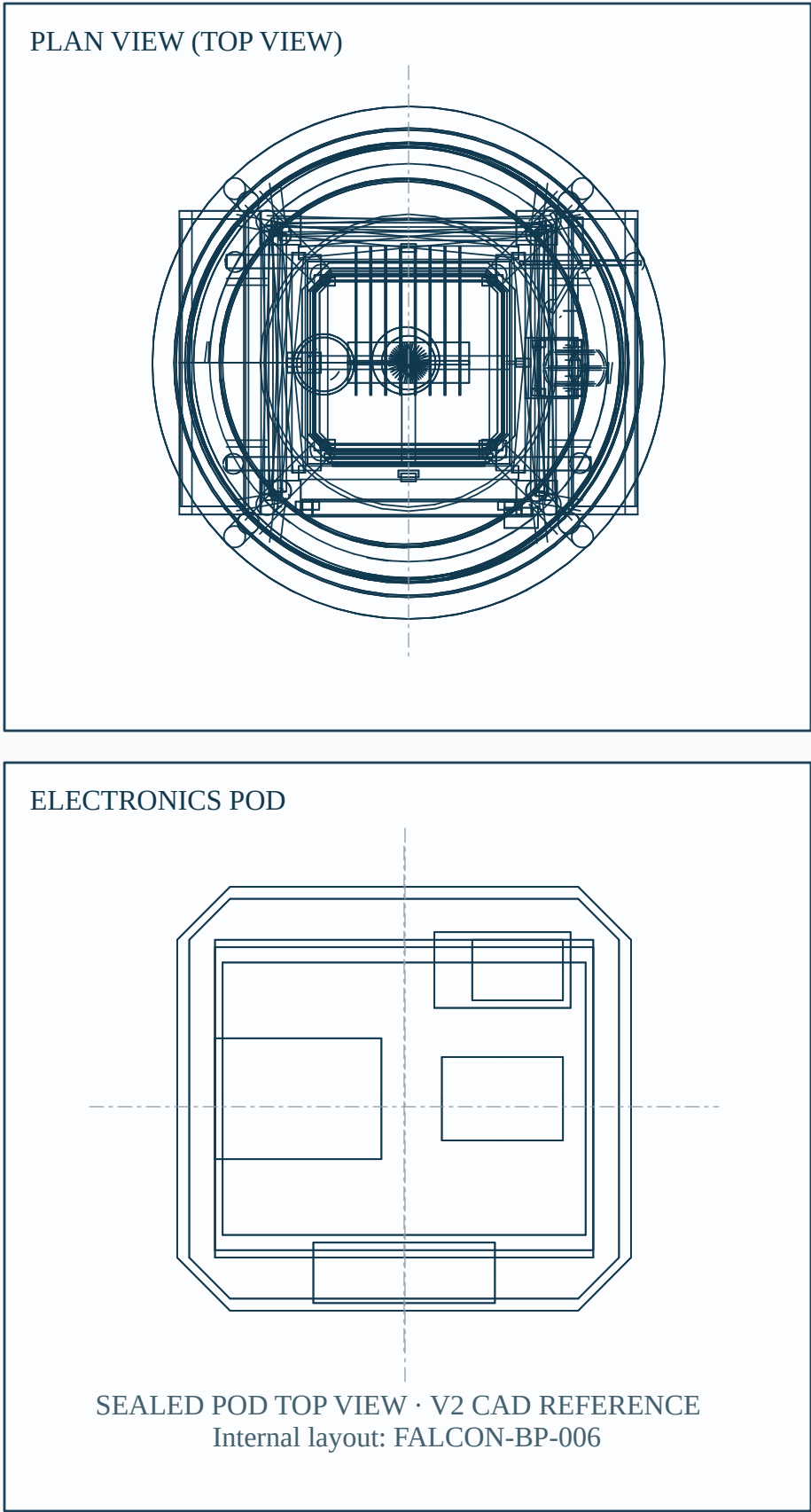
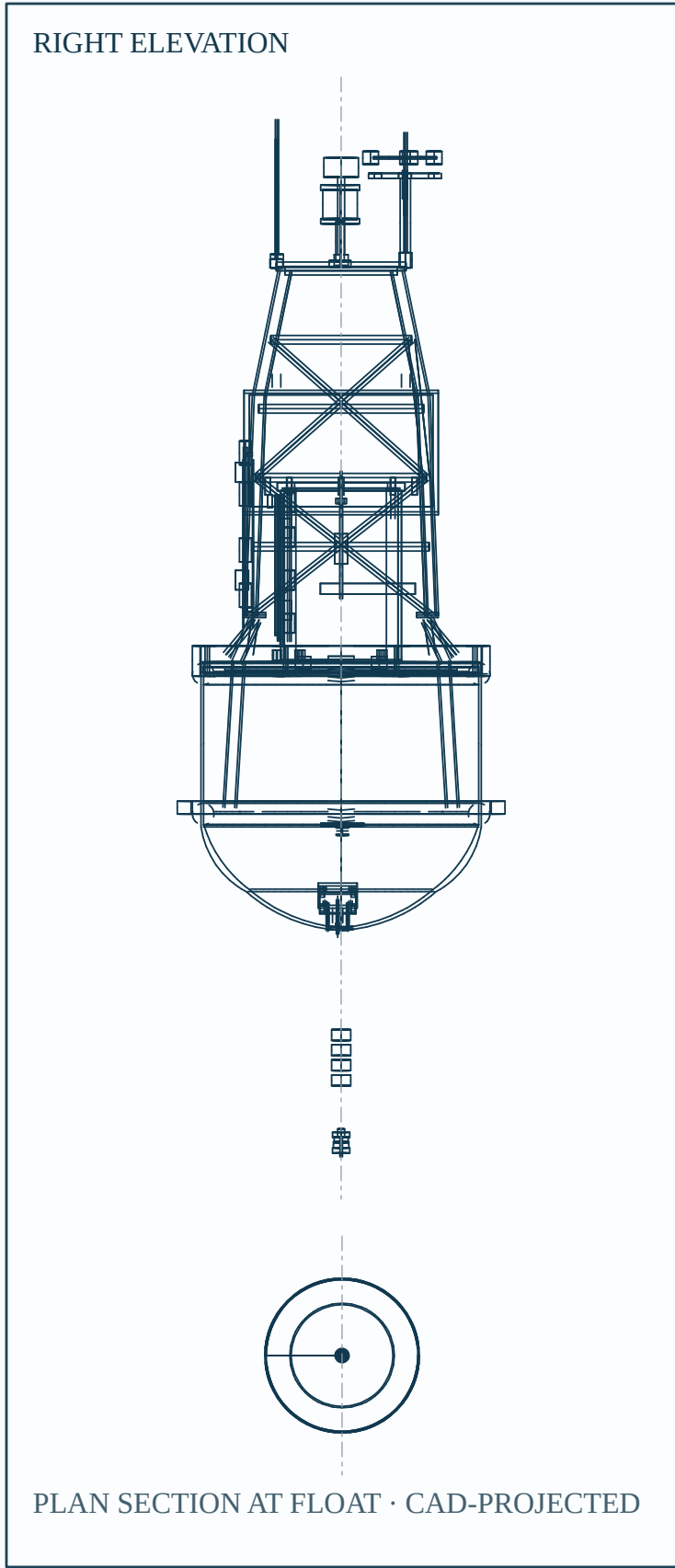
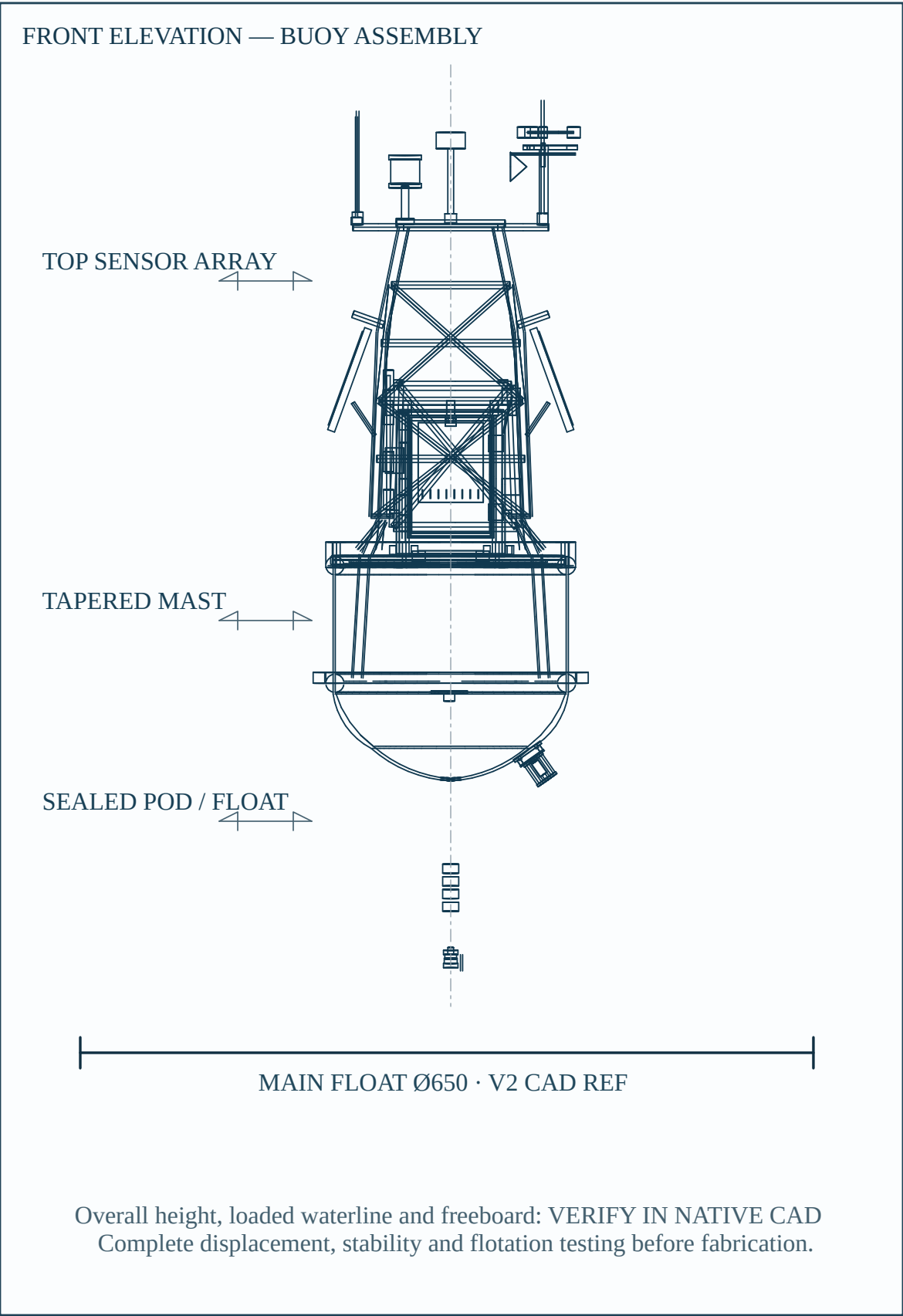
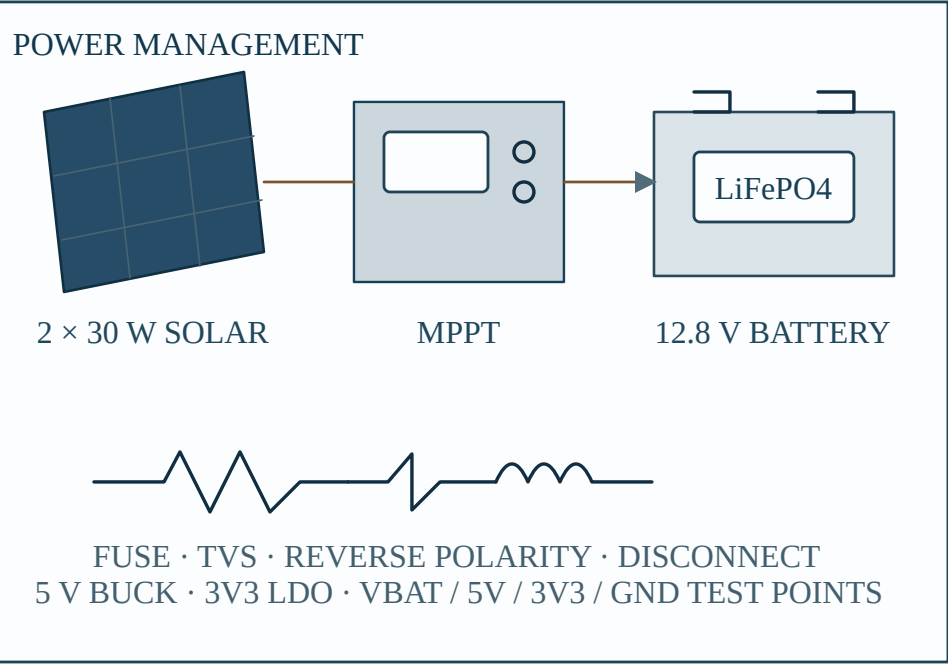


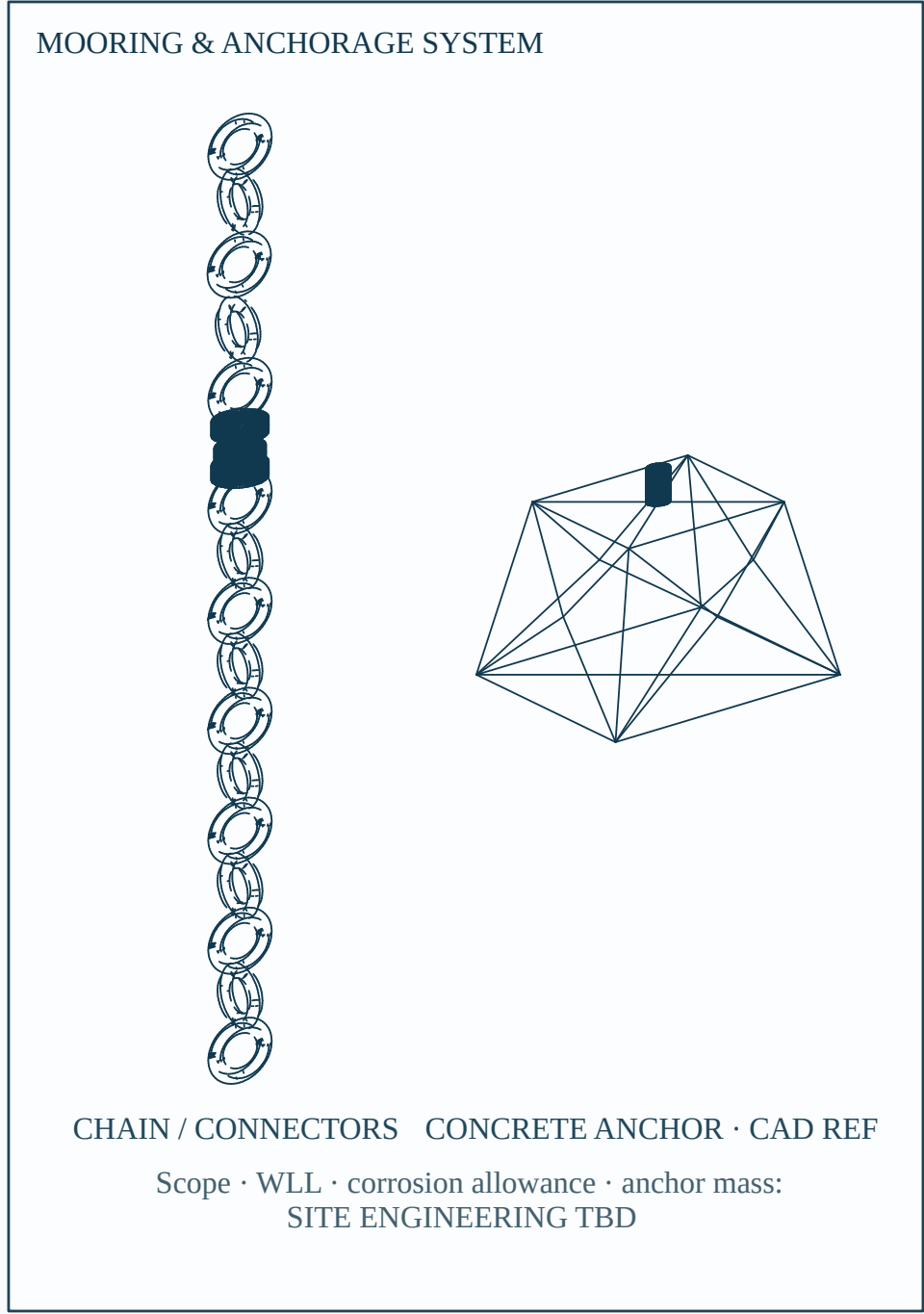


COMPONENT / INTERFACE NOTES	
S01 Bar02 R2	— pressure time series for estimated wave height; service limitation must be resolved.
S02 Ultimate GPS PID 746	— position/geofence; UART GPIO16/17; field scatter test required.
S03 SEN-15901	— wind pulse GPIO25 and vane through ADS1115 A0; marine durability unqualified.
S04 Celsius R2	— deployment water-temperature candidate; exact harness/address must be verified.
H01/H02 MCP9808 + INA260 ×2	— enclosure temperature and battery/solar electrical health.
C01/C02 ESP32-DevKitC V4 + SIM7600G-H candidate	— LTE interface and peak current not frozen.
EXCLUDED BNO085, load-cell/HX711, onboard Orange Pi and salinity sensor.	

GENERAL NOTES	
1.	CAD linework is projected from PROJECT-FALCON-V2.glb; do not scale this sheet.
2.	Project FALCON estimates wave height from submerged pressure time series at the shore Bay Station.
3.	Orange Pi / Bay Station computer remains ashore; no single-board computer is installed on the buoy.
4.	Verify purchased components, pin 1, voltages, connector orientation, sealing, loads and calibration.
5.	Complete flotation, stability, structure, thermal, ingress, corrosion, power and mooring reviews before build.
REFERENCE / NOT FOR FABRICATION	



WIRING SCHEMATIC OVERVIEW		
	ESP32	PROTECTED I/O
I2C SDA / SCL		
GPS UART RX / TX		
WIND PULSE / ADC		
LTE UART / USB		
3V3 / 5V / GND		
FUNCTIONAL OVERVIEW ONLY Use verified KiCad schematic and harness for fabrication		



DRAWING CONTROL		DRAWING TITLE ASSEMBLY & SUBSYSTEMS	
PROJECT FALCON-01			
DRAWING NO. FALCON-BP-000		REVISION	P3
SCALE NTS		DATE	2026-09-03
SOURCE V2 GLB + CONTROLLED DOCS		STATUS	REFERENCE / NOT FOR FABRICATION